

Decompression Sickness and Arterial Gas Embolism

Guideline Title: Decompression Sickness and
Arterial Gas Embolism

Guideline Number: GUID-3203-T004

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Department Numbers: 3203	

- CI. **PURPOSE:** Therapeutic goal: To recognize the patient with decompression illness and to provide and maintain adequate oxygenation, ventilation, and tissue perfusion. To provide adequate pain management and expeditious transport to the nearest appropriate facility. Decompression sickness may be caused during a scuba diving hazard or depressurization events in unpressurized aircraft.
- CII. **DEFINITIONS:**
- A. Decompression sickness – gas bubbles forming in venous blood or tissue causing localized joint pain (shoulders, elbows, knees, ankles) pruritus over upper torso, localized edema, and headache.
 - B. Arterial Gas Embolism – gas bubbles forming in arterial circulation causing neurologic (stroke like symptoms or paraplegia) or cardiovascular collapse
- CIII. **DEPARTMENT GUIDELINE:**
- A. Perform a general medical assessment and establish care per *GUID-G002 – General Patient Care*.
 - B. Hypothermia should be considered an aggravating factor in every aquatic accident. Remove wet clothing and protect against heat loss and wind chill.
 - C. Treatment at a hyperbaric chamber is a priority and will be arranged by the receiving facility, ideally a trauma center, after initial stabilization. Currently no Central Florida hyperbaric programs accept emergency cases.
 - D. Supplemental 100% oxygen by non-rebreather (NRB) should be given as soon as possible. Advanced airway techniques should be considered as the situation warrants.
 - E. Obtain IV/IO access.
 - F. In an effort to minimize any further gas expansion, the patient should be transported at 1000 feet or less if possible. In no case should the patient be transported at 3000 feet or higher.
 - G. Consider placing patient in mild Trendelenburg position to restore forward blood flow by placing the right ventricular outflow tract inferior to the right ventricular cavity, permitting air to migrate superiorly to a non-obstructing position.
 - H. Obtain as detailed a history as possible; in the case of a diving accident, the type of diving, depth, bottom time, number of dives in the past 72 hours, surface interval between repetitive dives over the past 72 hours, in-water decompression, site of diving

and environmental conditions, primary diving activity, presence of pre-disposing factors, dive complications, pre and post dive activities, and the onset of symptoms.

CIV. **DOCUMENTATION:**

A. EMS Charts

CV. **REFERENCES:**

- A. Divers Alert Network Emergency Hotline (919) 648-9111 can be called to assist in determining the nearest available hyperbaric center if necessary.
- B. Dick, A. P. & Massey, E. W. (1985). Neurologic presentation of decompression sickness and air embolism in sport divers. *Neurology*, 35(5), 667.
- C. Tetzlaff, K., Shank, E.S., & Muth, C.M. (2003). Evaluation and management of decompression illness –an intensivist’s perspective. *Intensive Care Med*, 29(12), 2128.